Software Test Plan (STP) - Student Attendance Management System

Project: Student Attendance Management System
Version: 1.0
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Status: Pending

1. Introduction

Purpose: This document defines the test plan for the Student Attendance Management System v1.0. It outlines objectives, scope, strategy, resources, schedule, and responsibilities for testing.

Scope: Testing covers authentication, role-based access, class and roster management, attendance marking/correction, attendance history, percentage calculation, low-attendance identification, report generation/export, audit logging, performance, security, recovery, and usability. Database engine internals and external institutional systems are excluded unless integrated through defined interfaces.

References: Student Attendance SRS v1.0, Student Attendance SAD v1.0.
Definitions: SAM (Student Attendance Management), SRS, SAD, STP, RTM, RBAC.

2. Test Items

- Authentication and authorization module
- Student and faculty account management
- Course, class, and enrollment management
- Attendance marking and correction module
- Attendance history and percentage calculation
- Low-attendance identification
- Attendance reporting and CSV export
- Dashboard and audit logging

3. Features to be Tested

Features mapped to SRS requirement IDs:
- SAM-F-001..SAM-F-002: Authentication and role-based access
- SAM-F-003..SAM-F-007: Attendance capture, duplicate prevention, and correction
- SAM-F-008..SAM-F-010: History, percentage, and low-attendance checks
- SAM-F-011..SAM-F-013: Report generation, filtering, and CSV export
- SAM-F-014..SAM-F-017: Administration, audit trail, and dashboard
- SAM-NF-001..SAM-NF-006: Performance, availability, security, recovery, usability, and retention
- SAM-SR-001..SAM-SR-005: Security requirements

4. Features Not to be Tested

- Internal implementation details of the database engine
- External institutional ERP functionality not exposed through the project API
- Physical biometric hardware, if considered as a future enhancement
- Browser/operating-system internals outside the application boundary

5. Test Approach / Strategy

Levels:
- Unit tests (service/module level)
- Integration tests (UI/API/database)
- System tests (end-to-end attendance workflows)
- Acceptance tests (UAT)

Types:
- Functional testing (attendance and administration features)
- Regression testing
- Performance testing (response time and concurrent users)
- Security testing (authentication, authorization, input validation)
- Recovery/data-integrity testing
- Usability testing (clarity and responsive UI)

Entry Criteria: Stable build delivered, test data available, test environment ready, and database schema migrated successfully.
Exit Criteria: 100% of planned test cases executed, 0 critical defects open, all high-priority acceptance criteria satisfied, and RTM coverage complete.

5.1 Security Validation

- Validate password hashing and absence of plaintext passwords in logs
- HTTPS/TLS verification
- Role-based authorization checks
- Unauthorized access attempts to another student's attendance
- Audit logging for attendance corrections and admin actions
- Input validation and API security testing

6. Test Environment

Hardware: Standard laptop/desktop and mobile browser for responsive checks.
Software: Student Attendance web application v1.0, Node.js/Express API, MySQL test database, modern Chrome/Firefox/Edge browser.
Tools: Browser developer tools, Postman (API), JMeter or equivalent (performance), test runner, and issue tracker.
Test Data: Synthetic students, faculty, classes, enrollments, attendance records, invalid credentials, duplicate submissions, and low-attendance scenarios.

7. Test Schedule

Milestones:

- Test case design: 02-Oct-2026
- Environment setup: 03-Oct-2026
- Test execution start: 04-Oct-2026
- Test execution end: 10-Oct-2026
- UAT: 11-Oct-2026 to 13-Oct-2026

8. Test Deliverables

- Test Plan (this document)
- Test Cases (manual & automated)
- Test Scripts
- Test Data
- Test Execution Logs
- Defect Reports
- Test Summary Report

9. Roles and Responsibilities

Role	Name	Responsibility
QA Lead	sanjit	Prepare plan, coordinate execution
Test Engineer	Chandraraj Patil	Design & execute test cases, log defects
Developer	Greeshma	Support defect fixes and triage
Product Owner / Instructor	Chethan BJ	Review test results and sign-off readiness

10. Risks and Mitigation

Risk	Mitigation
Delay in stable build delivery	Request early smoke builds from development team
Test environment downtime	Maintain backup test environment and database backup
Incomplete or inconsistent test data	Use controlled synthetic data and validate database fixtures
Unauthorized data exposure during testing	Use synthetic records and restrict test environment access

11. Assumptions & Dependencies

- Test database and application environment will be available before execution
- Synthetic student, faculty, class, enrollment, and attendance data will be provided
- Attendance policy threshold will be configured before report testing
- Browser versions required for testing will be available

12. Suspension & Resumption Criteria

Suspend testing if:

- Environment unavailable for >4 hours
- Build is too unstable and blocks >30% of planned test cases
- Database corruption prevents reliable execution

Resume testing if:

- Blocking defects are resolved
- Environment is stabilized and test data is restored
- Database consistency is verified

13. Test Case Management & Traceability

RTM ensures mapping of SRS requirements to test cases. The following test-case catalogue defines every test case referenced by the SRS RTM.

TC-AUTH-01: Valid login with registered credentials creates an authenticated session.

TC-AUTH-02: Role-based access prevents users from opening unauthorized functions.

TC-ATT-01: Faculty selects an assigned class/date and the correct enrolled roster is displayed.

TC-ATT-02: Faculty marks each enrolled student Present, Absent, or Late and submits the attendance.

TC-ATT-03: A duplicate attendance record for the same student, class, and date is rejected or safely updated.

TC-ATT-04: Authorized faculty corrects an attendance record and an audit timestamp is recorded.

TC-STU-01: Student views only their own attendance history by course and date.

TC-REP-01: Attendance percentage is calculated correctly from stored attendance records.

TC-REP-02: Students below the configured attendance threshold are flagged.

TC-REP-03: A report for the selected class/course/date range contains correct counts and percentages.

TC-REP-04: Report filters return only records matching student/class/course/date criteria.

TC-REP-05: CSV export contains the same data shown in the generated report.

TC-ADM-01: Administrator creates, updates, and deactivates student/faculty accounts.

TC-ADM-02: Administrator creates classes/courses and changes enrollments reflected in rosters.

TC-AUD-01: Attendance corrections and administrative/security events record actor, action, target, and timestamp.

TC-DASH-01: Dashboard displays current attendance summaries and report-generation status.

TC-PERF-01: 95% of common attendance/report operations complete within 3 seconds under normal load.

TC-OPS-01: Availability monitoring demonstrates at least 99.5% monthly availability excluding scheduled maintenance.

TC-SEC-01: Authenticated application traffic uses HTTPS/TLS and no plaintext application traffic is accepted.

TC-REC-01: A valid backup restores attendance data without inconsistent records.

TC-UX-01: Responsive UI passes defined desktop and mobile viewport usability checks.

TC-AUD-02: Audit records remain retrievable for the configured retention period.

TC-SEC-02: Passwords are stored as salted one-way hashes and plaintext passwords are absent from logs.

TC-SEC-03: Every protected API operation enforces server-side role authorization.

TC-SEC-04: A student cannot access another student's attendance data through direct or modified requests.

14. Test Metrics & Reporting

Metrics collected:
- % test cases executed
- % passed/failed
- Defect density
- Defect aging
- Requirement coverage
- Average/95th-percentile response time

Reports:
- Daily execution status
- Defect summary
- Final Test Summary Report

15. Approvals

Role	Name	Signature / Date
QA Lead	Sanjit	
Dev Lead	Greeshma G	
Product Owner / Instructor	Chetah bj	